Python

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• Strings :-

→ concatenation (+) 'code' + 'chef'

→ 'code' * 3

→ len('code')

→ 'code'.upper()

→ 'code'.lower()

→ s = 'code'

s[0]

s[0:4]

jump ↓ s[0:4:2]

reverse slicing ↓ s[4:0:-1]

→ s = 'chaf'

new_s = s.replace('a', 'e')

→ String is immutable.

★ Input -

- int(input())
- multiple string input

  a, b, c = input().split()
- multiple integer input

  a, b, c = map(int, input().split())

★ List

0 1 2 → index

a = ["red", "green", "blue"] → list

a[0] ⇒ red

- list is ~~imm~~ mutable.
- a[0] = 'orange'
- len(a)
- a[0:3]
- a[3:0:-1]

fruits = ['apple', 'banana', 'cherry']

- **List Append** → add an item to end of list.

fruits.append("orange")

['apple', 'banana', 'cherry', 'orange']

- **List Insert** → add element to a list.

fruits.insert(1, "orange")

['apple', 'orange', 'banana', 'cherry']

- **List remove** → remove first occurence of the specified item of list.

fruits = ['apple', 'banana', 'cherry', 'banana']

fruits.remove('banana')

['apple', 'cherry', 'banana']

- **List pop** → takes last element

fruits = ['apple', 'banana', 'cherry']

fruits.pop()

['apple', banana']

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- List pop :- remove by giving index of item.

fruits = ['apple', 'banana', 'cherry']

fruits.pop(1) → removes banana

removed_fruits = fruits.pop(1) → remove cherry

fruits → ['apple']

- List concat

[1, 2] + [3, 4] = [1, 2, 3, 4]

- List String User Input

fruits = input().split()

- List Integer User Input

numbers = list(map(int, input().split()))

☆ Tuples

- immutable

my_tuple = (1, 2, 3)
a, b, c = my_tuple

len(my_tuple)
my_tuple.count(1)

☆ Dictionary

```
dict1 = {'name': 'Alice', 'age': 30}
dict1['name'] # Alice
dict1['location'] = 'new york'
```

- **Removing item**

  del dict1['name'] → remove specific key

- pop() → remove specific key & get corresponding value.

  removed_age = ~~my_d~~ dict1.pop('age') # 30

- clear(): remove all key value pairs

  dict1.clear()

- **Iterating through dictionary**

→ key

```
for key in dict1:
    print(key)
```

→ values

```
for value in dict1.values():
    print(value)
```

→ key - value

```
for key, value in dict1.items():
        print(key, value)
```

- Key existence

→ in :- used to check if key is present in dictionary.

```
~~k~~ "key" in dict1
```

→ get ( ): retrive ~~key~~ value for key.

```
dict1.get ("name")
```

- Nested Dict

```
nested_dict = {
    'person1': {'name': 'Alice', 'age': 30},
    'person2': {'name': 'Bob', 'age' = 25}
}

nested_dict ['person1'] ['name'] # 'Alice'
```